

Quantum Field Theory: Problem Set 1

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1 The BCS Ground State and the Gap Equation

Consider a system of spin- $\frac{1}{2}$ fermions described by the mean-field BCS Hamiltonian

$$H_{\text{BCS}} = \sum_{\mathbf{k}, \sigma} \xi_{\mathbf{k}} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \sum_{\mathbf{k}} \left(\Delta c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger + \Delta c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \right),$$

where

$$\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu,$$

and Δ is assumed to be real.

(a) Nambu form of the Hamiltonian

Define the Nambu spinor

$$\Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix}.$$

Show that the Hamiltonian may be written as

$$H_{\text{BCS}} = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger \begin{pmatrix} \xi_{\mathbf{k}} & -\Delta \\ -\Delta & -\xi_{\mathbf{k}} \end{pmatrix} \Psi_{\mathbf{k}} + \text{constant}.$$

Identify the constant term.

(b) Bogoliubov transformation

Introduce new fermionic operators

$$\gamma_{\mathbf{k}\uparrow} = u_k c_{\mathbf{k}\uparrow} - v_k c_{-\mathbf{k}\downarrow}^\dagger,$$

$$\gamma_{-\mathbf{k}\downarrow} = u_k c_{-\mathbf{k}\downarrow} + v_k c_{\mathbf{k}\uparrow}^\dagger,$$

where u_k and v_k are real coefficients.

1. Show that the operators γ satisfy canonical fermionic anticommutation relations provided

$$u_k^2 + v_k^2 = 1.$$

2. Determine u_k and v_k such that the Hamiltonian becomes diagonal in terms of the quasiparticle operators.

(c) Quasiparticle spectrum

Show that the diagonalized Hamiltonian can be written as

$$H_{\text{BCS}} = E_0 + \sum_{\mathbf{k}} E_k \left(\gamma_{\mathbf{k}\uparrow}^\dagger \gamma_{\mathbf{k}\uparrow} + \gamma_{-\mathbf{k}\downarrow}^\dagger \gamma_{-\mathbf{k}\downarrow} \right),$$

where

$$E_k = \sqrt{\xi_k^2 + \Delta^2}.$$

Show that

$$u_k^2 = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right), \quad v_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right).$$

(d) Construction of the BCS ground state

Let $|\text{BCS}\rangle$ denote the vacuum of the Bogoliubov quasiparticles:

$$\gamma_{\mathbf{k}\uparrow} |\text{BCS}\rangle = 0, \quad \gamma_{-\mathbf{k}\downarrow} |\text{BCS}\rangle = 0.$$

For a fixed momentum pair $(\mathbf{k}, -\mathbf{k})$ consider the ansatz

$$|\psi_k\rangle = A_k |0\rangle + B_k c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger |0\rangle.$$

Using the quasiparticle vacuum conditions, show that

$$A_k = u_k, \quad B_k = v_k.$$

Hence prove that

$$|\text{BCS}\rangle = \prod_{\mathbf{k}} \left(u_k + v_k c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \right) |0\rangle.$$

(e) Pair condensate

Compute the expectation value

$$\langle \text{BCS} | c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} | \text{BCS} \rangle.$$

Show that

$$\langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle = u_k v_k = \frac{\Delta}{2E_k}.$$

Explain why a nonzero value of this quantity signals Cooper-pair condensation.

(f) Particle number

Let

$$N = \sum_{\mathbf{k}, \sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}.$$

Show that

$$\langle N \rangle = 2 \sum_{\mathbf{k}} v_{\mathbf{k}}^2.$$

Explain why the BCS ground state is not an eigenstate of the particle-number operator.

(g) Self-consistent gap equation

Suppose the microscopic interaction is

$$H_{\text{int}} = -g \sum_{\mathbf{k}, \mathbf{k}'} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}, \quad g > 0.$$

In the mean-field approximation, define the gap parameter by

$$\Delta = g \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle.$$

Using the result of part (e), show that

$$\Delta = g \sum_{\mathbf{k}} u_{\mathbf{k}} v_{\mathbf{k}}.$$

Hence derive the self-consistency condition

$$1 = g \sum_{\mathbf{k}} \frac{1}{2E_{\mathbf{k}}},$$

where

$$E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}.$$

This equation is known as the *BCS gap equation*.

(h) Continuum limit

Assume that pairing occurs only within a thin shell around the Fermi surface:

$$|\xi| < \omega_D,$$

where ω_D is the Debye frequency.

Assume further that the density of states is approximately constant near the Fermi surface:

$$N(\xi) \simeq N(0).$$

Convert the momentum sum into an integral and show that the gap equation becomes

$$1 = gN(0) \int_0^{\omega_D} \frac{d\xi}{\sqrt{\xi^2 + \Delta^2}}.$$

Evaluate the integral explicitly and obtain

$$1 = gN(0) \operatorname{arcsinh}\left(\frac{\omega_D}{\Delta}\right).$$

(i) Weak-coupling solution

Assume that

$$\Delta \ll \omega_D.$$

Using

$$\operatorname{arcsinh}(x) \simeq \ln(2x), \quad x \gg 1,$$

show that the gap equation reduces to

$$1 = gN(0) \ln\left(\frac{2\omega_D}{\Delta}\right).$$

Hence derive the celebrated weak-coupling result

$$\Delta = 2\omega_D \exp\left[-\frac{1}{gN(0)}\right].$$

(j) Interpretation

Answer the following questions.

1. Why does the superconducting gap depend nonperturbatively on the coupling constant g ?
2. Explain why the result of part (i) implies that an arbitrarily weak attractive interaction leads to a superconducting instability.
3. Compare the dependence

$$\Delta \sim e^{-1/(gN(0))}$$

with other examples of nonperturbative mass generation encountered in quantum field theory.

2 Bogoliubov Theory of a Weakly Interacting Bose Gas

Consider a system of bosons of mass m confined to a volume V and interacting through a contact interaction. The Hamiltonian is

$$H = \sum_{\mathbf{k}} \varepsilon_{\mathbf{k}} a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}} + \frac{g}{2V} \sum_{\mathbf{k}, \mathbf{p}, \mathbf{q}} a_{\mathbf{k}+\mathbf{q}}^{\dagger} a_{\mathbf{p}-\mathbf{q}}^{\dagger} a_{\mathbf{p}} a_{\mathbf{k}},$$

where

$$\varepsilon_k = \frac{k^2}{2m}, \quad g > 0.$$

Assume that the temperature is zero and that a macroscopic number of particles occupies the zero-momentum state.

(a) Bose-Einstein condensate and the Bogoliubov approximation

Let N_0 denote the number of particles in the $\mathbf{k} = 0$ mode and define the condensate density

$$n_0 = \frac{N_0}{V}.$$

1. Explain why, when $N_0 \gg 1$, one may approximate

$$a_0, a_0^\dagger \rightarrow \sqrt{N_0}.$$

2. Keeping only terms up to quadratic order in the operators $a_{\mathbf{k}}$ and $a_{\mathbf{k}}^\dagger$ with $\mathbf{k} \neq 0$, show that the Hamiltonian becomes

$$H = E_{\text{MF}} + \sum_{\mathbf{k} \neq 0} \left[(\varepsilon_k + gn_0) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{gn_0}{2} (a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + a_{-\mathbf{k}} a_{\mathbf{k}}) \right].$$

3. Determine the mean-field energy E_{MF} .

(b) Nambu representation

Define the Nambu spinor

$$\Phi_{\mathbf{k}} = \begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix}.$$

Show that the quadratic Hamiltonian may be written in the form

$$H = E_{\text{MF}} + \frac{1}{2} \sum_{\mathbf{k} \neq 0} \Phi_{\mathbf{k}}^\dagger \begin{pmatrix} \varepsilon_k + gn_0 & gn_0 \\ gn_0 & \varepsilon_k + gn_0 \end{pmatrix} \Phi_{\mathbf{k}} + \text{constant}.$$

Discuss briefly how this matrix differs from the corresponding matrix in BCS theory.

(c) Bosonic Bogoliubov transformation

Introduce new operators

$$b_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger,$$

$$b_{\mathbf{k}}^\dagger = u_k a_{\mathbf{k}}^\dagger + v_k a_{-\mathbf{k}},$$

where u_k and v_k are real coefficients.

1. Show that preservation of the canonical bosonic commutation relations requires

$$u_k^2 - v_k^2 = 1.$$

2. Explain why this condition differs from the corresponding fermionic relation encountered in BCS theory.

(d) Diagonalization of the Hamiltonian

Determine u_k and v_k such that the Hamiltonian becomes diagonal in terms of the quasiparticle operators $b_{\mathbf{k}}$.

Show that the diagonalized Hamiltonian can be written as

$$H = E_0 + \sum_{\mathbf{k} \neq 0} E_k b_{\mathbf{k}}^\dagger b_{\mathbf{k}},$$

where

$$E_k = \sqrt{\varepsilon_k(\varepsilon_k + 2gn_0)}.$$

Determine the ground-state energy shift E_0 .

(e) Bogoliubov amplitudes

Show that

$$u_k^2 = \frac{1}{2} \left(\frac{\varepsilon_k + gn_0}{E_k} + 1 \right),$$

$$v_k^2 = \frac{1}{2} \left(\frac{\varepsilon_k + gn_0}{E_k} - 1 \right).$$

Verify explicitly that

$$u_k^2 - v_k^2 = 1.$$

Study the behavior of u_k and v_k in the limits

$$k \rightarrow 0, \quad k \rightarrow \infty.$$

(f) Bogoliubov vacuum

The Bogoliubov ground state $|\text{Bog}\rangle$ is defined by

$$b_{\mathbf{k}}|\text{Bog}\rangle = 0 \quad \text{for all } \mathbf{k} \neq 0.$$

For a fixed momentum pair $(\mathbf{k}, -\mathbf{k})$ consider the ansatz

$$|\psi_k\rangle = \sum_{n=0}^{\infty} C_n |n_k, n_{-k}\rangle.$$

1. Use the condition

$$b_{\mathbf{k}}|\psi_k\rangle = 0$$

to derive a recursion relation for the coefficients C_n .

2. Solve the recursion relation and show that

$$|\psi_k\rangle = \frac{1}{u_k} \sum_{n=0}^{\infty} \left(-\frac{v_k}{u_k} \right)^n |n_k, n_{-k}\rangle.$$

3. Explain why the Bogoliubov ground state is a two-mode squeezed vacuum.

(g) Condensate depletion

The number of particles outside the condensate is

$$N_{\text{ex}} = \sum_{\mathbf{k} \neq 0} a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}}.$$

1. Show that in the Bogoliubov ground state

$$\langle N_{\text{ex}} \rangle = \sum_{\mathbf{k} \neq 0} v_k^2.$$

2. Explain physically why interactions deplete the condensate even at zero temperature.

(h) Emergence of superfluid sound

Study the low-momentum limit of the Bogoliubov spectrum.

Show that for

$$k \ll \sqrt{2mgn_0},$$

the excitation energy becomes linear in momentum:

$$E_k \simeq c_s k.$$

Determine the sound velocity

$$c_s = \sqrt{\frac{gn_0}{m}}.$$

Explain why the low-energy excitations are called phonons.

(i) Landau criterion for superfluidity

The Landau critical velocity is defined by

$$v_c = \min_k \frac{E_k}{k}.$$

Using the Bogoliubov dispersion relation,

$$E_k = \sqrt{\varepsilon_k(\varepsilon_k + 2gn_0)},$$

show that

$$v_c = c_s.$$

Explain why the existence of a nonzero critical velocity implies dissipationless flow at sufficiently small velocities.

(j) Goldstone mode and spontaneous symmetry breaking

The microscopic Hamiltonian is invariant under the global $U(1)$ transformation

$$a_{\mathbf{k}} \rightarrow e^{i\alpha} a_{\mathbf{k}}.$$

Answer the following questions.

1. Why does the condensate select a definite phase?
2. Why is the Bogoliubov excitation spectrum gapless?
3. Explain how the phonon mode may be interpreted as the Goldstone boson associated with spontaneous breaking of the global $U(1)$ symmetry.
4. Compare the role of the phonon in the weakly interacting Bose gas with the role of the Cooper-pair condensate in BCS theory.

3 Bunching and Anti-Bunching Effects: Hanbury Brown–Twiss Interferometry

One of the most direct manifestations of quantum statistics is found in density-density correlation functions. In this problem, you will study how fermionic and bosonic quantum statistics produce qualitatively different spatial correlations. Fermions exhibit *anti-bunching*, while bosons exhibit *bunching*. These effects form the basis of Hanbury Brown–Twiss (HBT) interferometry and have been observed in systems ranging from photons and ultracold atoms to electrons.

3.1 Fermionic Anti-Bunching in a Fermi Liquid

Consider a homogeneous system of spinless noninteracting fermions at zero temperature in a volume V .

The field operator is

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\mathbf{k}},$$

where

$$\{c_{\mathbf{k}}, c_{\mathbf{p}}^\dagger\} = \delta_{\mathbf{k}\mathbf{p}}.$$

The density operator is

$$n(\mathbf{r}) = \psi^\dagger(\mathbf{r})\psi(\mathbf{r}).$$

The ground state is the filled Fermi sea

$$|FS\rangle = \prod_{|\mathbf{k}| < k_F} c_{\mathbf{k}}^\dagger |0\rangle.$$

(a) One-body density matrix

Compute

$$G(\mathbf{r}) = \langle \psi^\dagger(\mathbf{r}) \psi(0) \rangle$$

in the Fermi-sea ground state and show that

$$G(\mathbf{r}) = \int_{|\mathbf{k}| < k_F} \frac{d^3k}{(2\pi)^3} e^{-i\mathbf{k}\cdot\mathbf{r}}.$$

(b) Density-density correlation function

Define

$$C(\mathbf{r}) = \langle n(\mathbf{r}) n(0) \rangle.$$

Using only the canonical anticommutation relations and the occupation numbers

$$\langle c_{\mathbf{k}}^\dagger c_{\mathbf{p}} \rangle = \delta_{\mathbf{k}\mathbf{p}} \Theta(k_F - k),$$

show that

$$C(\mathbf{r}) = n^2 - |G(\mathbf{r})|^2 + n \delta^{(3)}(\mathbf{r}).$$

Hint: First rewrite

$$\psi(\mathbf{r}) \psi^\dagger(0) = \delta^{(3)}(\mathbf{r}) - \psi^\dagger(0) \psi(\mathbf{r}).$$

Explain the origin of the minus sign.

(c) Pair correlation function

Define

$$g^{(2)}(\mathbf{r}) = \frac{\langle n(\mathbf{r}) n(0) \rangle}{n^2}, \quad r \neq 0.$$

Show that

$$g^{(2)}(\mathbf{r}) = 1 - \left| \frac{G(\mathbf{r})}{n} \right|^2.$$

(d) The Fermi hole

Show that

$$g^{(2)}(0) = 0.$$

Explain physically why identical fermions avoid one another even in the absence of interactions.

(e) Explicit evaluation

Evaluate

$$G(r) = \frac{1}{2\pi^2 r} \int_0^{k_F} k \sin(kr) dk$$

and show that

$$G(r) = n 3 \frac{\sin(k_F r) - k_F r \cos(k_F r)}{(k_F r)^3}.$$

Hence derive

$$g^{(2)}(r) = 1 - 9 \left[\frac{\sin(k_F r) - k_F r \cos(k_F r)}{(k_F r)^3} \right]^2.$$

Sketch the function $g^{(2)}(r)$ and identify the characteristic length scale governing the anti-bunching effect.

(f) Static structure factor

Define

$$\rho_{\mathbf{q}} = \sum_{\mathbf{k}} c_{\mathbf{k}+\mathbf{q}}^\dagger c_{\mathbf{k}}$$

and

$$S(\mathbf{q}) = \frac{1}{N} \langle \rho_{\mathbf{q}} \rho_{-\mathbf{q}} \rangle.$$

Show that

$$S(q) \rightarrow 0 \quad (q \rightarrow 0).$$

Explain why long-wavelength density fluctuations are suppressed in a Fermi liquid.

3.2 Bosonic Bunching and the Hanbury Brown–Twiss Effect

Consider a homogeneous gas of noninteracting bosons in thermal equilibrium.

The field operator is

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} a_{\mathbf{k}},$$

where

$$[a_{\mathbf{k}}, a_{\mathbf{p}}^\dagger] = \delta_{\mathbf{k}\mathbf{p}}.$$

The density operator is again

$$n(\mathbf{r}) = \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}).$$

(a) One-body density matrix

Define

$$G(\mathbf{r}) = \langle \psi^\dagger(\mathbf{r}) \psi(0) \rangle.$$

Show that

$$G(\mathbf{r}) = \int \frac{d^3k}{(2\pi)^3} n_B(\epsilon_k) e^{-i\mathbf{k}\cdot\mathbf{r}},$$

where

$$n_B(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} - 1}.$$

(b) Density-density correlation function

Compute

$$C(\mathbf{r}) = \langle n(\mathbf{r}) n(0) \rangle.$$

Using only the canonical commutation relations and the thermal occupation numbers

$$\langle a_{\mathbf{k}}^\dagger a_{\mathbf{p}} \rangle = \delta_{\mathbf{k}\mathbf{p}} n_B(\epsilon_k),$$

show that

$$C(\mathbf{r}) = n^2 + |G(\mathbf{r})|^2 + n \delta^{(3)}(\mathbf{r}).$$

Hint: Use

$$\psi(\mathbf{r}) \psi^\dagger(0) = \delta^{(3)}(\mathbf{r}) + \psi^\dagger(0) \psi(\mathbf{r}).$$

Explain why the sign differs from the fermionic result.

(c) Pair correlation function

Show that

$$g^{(2)}(\mathbf{r}) = 1 + \left| \frac{G(\mathbf{r})}{n} \right|^2.$$

(d) Bunching

Show that

$$g^{(2)}(0) = 2.$$

Explain the physical significance of this result.

How does it illustrate the tendency of bosons to bunch together?

(e) Bose–Einstein condensate

Suppose a Bose–Einstein condensate forms with condensate density n_0 .

Write

$$\psi(\mathbf{r}) = \sqrt{n_0} + \delta\psi(\mathbf{r}).$$

Assuming zero temperature and neglecting quantum depletion, show that

$$g^{(2)}(0) = 1.$$

Explain why a coherent condensate exhibits Poissonian number fluctuations rather than thermal bunching.

(f) Static structure factor

Define

$$\rho_{\mathbf{q}} = \sum_{\mathbf{k}} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}}$$

and

$$S(\mathbf{q}) = \frac{1}{N} \langle \rho_{\mathbf{q}} \rho_{-\mathbf{q}} \rangle.$$

Show that for an ideal Bose gas, the long-wavelength density fluctuations are enhanced relative to the fermionic case.

Discuss the physical origin of this enhancement.

Discussion

Compare the fermionic and bosonic results and summarize how quantum statistics manifest themselves through the sign of the exchange contribution:

$$\langle n(\mathbf{r})n(0) \rangle = n^2 \pm |G(\mathbf{r})|^2.$$

Explain how this single sign difference leads to

- anti-bunching of fermions,
- bunching of bosons,
- suppression of density fluctuations in Fermi systems,
- enhancement of density fluctuations in Bose systems,
- the Hanbury Brown–Twiss effect.